Quantification: 701_z36c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	3
interior lacuna count	3
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 3)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	883.667	804	491.368	px^2
major_axis_length_px	lacuna shape	42.365	41.5213	11.3585	px
minor_axis_length_px	lacuna shape	28.8076	33.4623	9.1056	px
aspect_ratio	lacuna shape	1.5067	1.5622	0.243	unitless
eccentricity	lacuna shape	0.7244	0.7683	0.1168	unitless
solidity	lacuna shape	0.8359	0.8918	0.123	unitless
roots_count	headline	7	7	1	count
ring_length_r30_px	headline	368	366	74.0203	px
ring_length_r60_px	reported	1069	1091	156.167	px
owned_length_px	ownership-dependent	2016.77	1330.54	1685.16	px
edge_count	ownership-dependent	151.333	91	142.43	count
mean_edge_length_px	ownership-dependent	14.3794	14.6214	1.7336	px
perimeter_px	normalised	160.123	158.267	66.6027	px
ring_area_r30_px2	normalised	6434	6849	1035.85	px^2
ring_area_r60_px2	normalised	18434.3	19116	2030.22	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0572218	0.056137	0.00698027	px^-1
ring_density_r60	normalised	0.0579389	0.0559098	0.00483216	px^-1
roots_per_100px_perimeter	normalised	4.9051	3.7911	2.1736	per 100 px
ring_attached_length_r30_px	network v2	281.333	278	89.0468	px
ring_attached_length_r60_px	network v2	572.667	514	286.54	px
ring_length_w_r30_px	network v2	420.457	408.291	82.7506	px
ring_length_w_r60_px	network v2	1202.67	1215.15	179.672	px
sholl_crossings_r10	network v2	9	10	2.6458	count
sholl_crossings_r20	network v2	8	9	1.7321	count
sholl_crossings_r30	network v2	14.6667	17	4.9329	count
loop_count	loops, report only	8.6667	3	10.6927	count
loop_edge_share	loops, report only	0.2209	0.1758	0.1078	share
ring_width_mean_r30_px	width	5.8565	5.8682	1.103	px

Per field
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measure	value	units
analysed_area_px2	767286	px^2
total_skeleton_length_px	43259	px
canalicular_length_density_per_px	0.0563792	px^-1
junction_count	1301	count
junction_density_per_px2	0.00169559	px^-2
lacuna_count	3	count

lacunae_per_px2	3.91e-06	px^-2
skeleton_component_count	720	count
mean_component_length_px	60.0819	px
median_component_length_px	11	px
field_length_density_w_per_px	0.0633271	px^-1
field_density_without_flagged_per_px	0.0400054	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	64	px
unattached_component_count	5	count
canalicular_length_density_attached_per_px	0.0563792	px^-1
field_length_density_with_unattached_per_px	0.0564627	px^-1
excluded_area_px2	278639	px^2
analysed_field_area_px2	769937	px^2
loop_count	173	count
loop_edge_share	0.1969	share
edge_count	4211	count

Canalicular width, the new measure

measure	value	units
width_mean_px	5.6866	px
width_median_px	5.6569	px
width_p10_px	2	px
width_p90_px	8.9443	px
width_pixels_used	35934	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	804	41.5213	33.4623	1.2408	0.592	0.6949	-0.5169	176.293	670.124	False
2	437	31.4519	18.3155	1.7172	0.8129	0.8918	1.4887	285.545	929.915	False
3	1410	54.1218	34.645	1.5622	0.7683	0.921	-1.5551	413.697	828.063	False

Network per lacuna

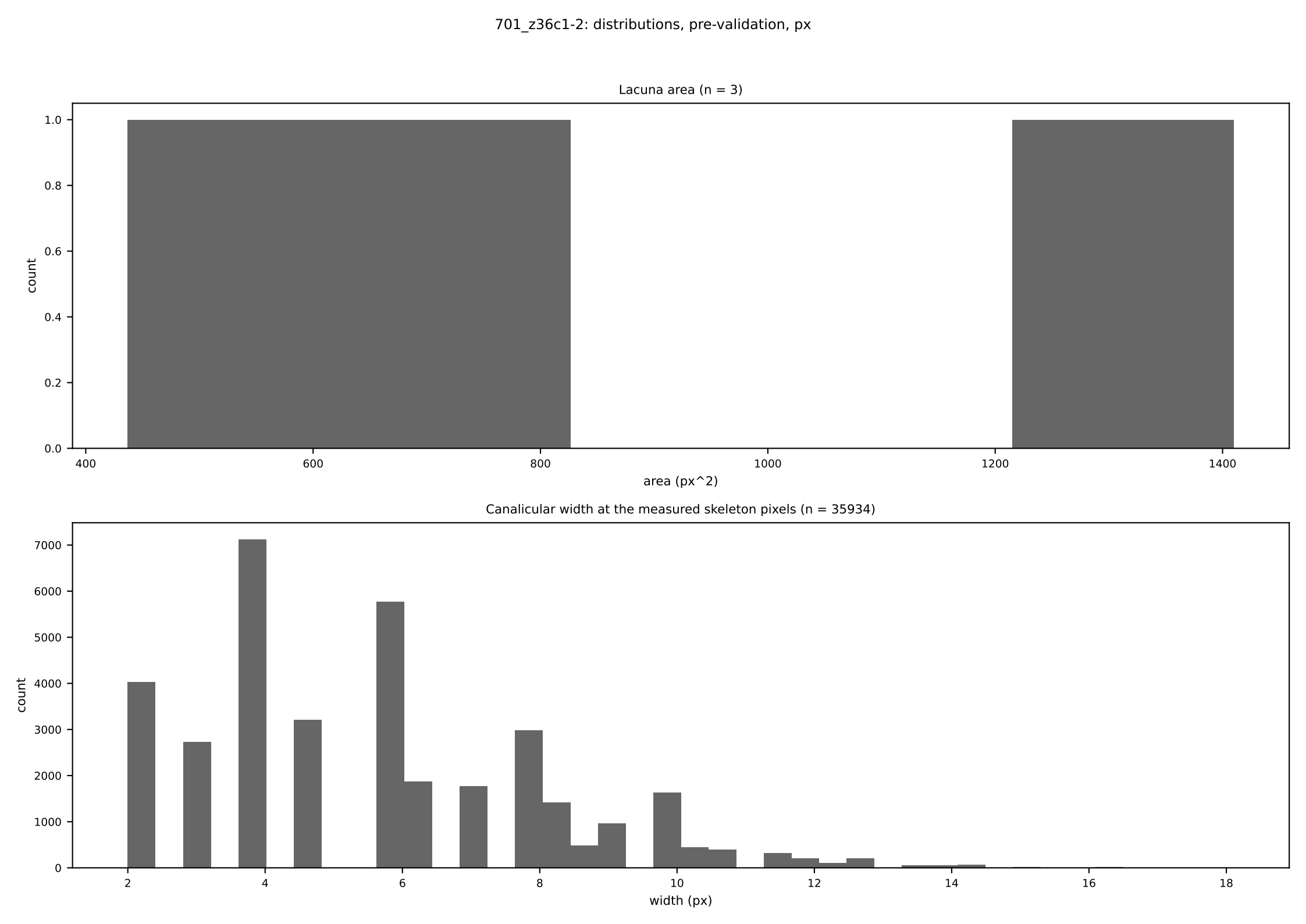
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	443	1213	3936.78	314	12.5375	5.8682
2	7	295	903	1330.54	91	14.6214	6.9536
3	6	366	1091	782.985	49	15.9793	4.7476

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	227.635	6849	19116	1	1	0.064681	0.0634547	3.5144
2	94.468	5255	16151	1	1	0.056137	0.0559098	7.4099
3	158.267	7198	20036	1	1	0.0508475	0.054452	3.7911

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	372	884	508.617	1375.79	11	9	18
2	278	514	344.463	1017.09	6	9	9
3	194	320	408.291	1215.15	10	6	17


701_z36c1-2: distributions, pre-validation, px
Lacuna area (n = 3)
count
area (px^2)
0.0
0.2
0.4
0.6
0.8
1.0
400
600
800
1000
1200
1400
Canalicular width at the measured skeleton pixels (n = 35934)
count
width (px)
0
1000
2000
3000
4000
5000
6000
7000
2
4
6
8
10
12
14
16
18